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Group 24 Project Report

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Section 1: Team Roles & Responsibilities

Team Member	Responsibilities
Hoo Xin Yan	<u>Apparel and Aerospace Stocks:</u> - Nike, Inc. (NKE) - The Boeing Company (BA) Developed MLP Classifier Strategy
Joshua Wong	<u>Commodity Stocks:</u> - Exxon Mobil Corporation (XOM) - Chevron Corporation (CVX) - Caterpillar (CAT) Developed Random Forest Strategy
Muhammad Ali Imran	<u>Technology Stocks:</u> - Apple Inc. (AAPL) - Microsoft Corporation (MSFT) - Alphabet Inc. (GOOG) Developed Linear Regression Strategy
Ong Sing May	<u>Healthcare and Semiconductors Stocks:</u> - Johnson & Johnson (JNJ) - UnitedHealth Group Incorporated (UNH) - NVIDIA Corporation (NVIDIA) Developed Decision Tree Strategy
All members	Collaborated on Kelly Criterion implementation, benchmark return calculations, and final integration of the trading strategies across different sectors
All members	Contributed to presentation slides, report writing, visualisation generation, and final code polishing to ensure consistency and clarity throughout the project

Section 2: Stock Selection & Strategies

As a group, we were provided with 20 stocks and chose to categorise them by sector, for example, commodity-related stocks such as ExxonMobil, Chevron, and Caterpillar, or technology-related stocks such as Apple, Microsoft, and Google. We reasoned that stocks within the same sector often exhibit similar price behaviors and patterns, making it more practical to develop trading strategies that could be applied across multiple stocks within the same category.

Our selection of sectors and stocks were not arbitrary. Instead, we applied both quantitative and qualitative reasoning. For example, qualitatively, we considered the availability of external data that may be of high relevance for a particular stock or sector. Quantitatively, we made use of numerical values, such as the realised outperformance of our relevant strategies with the benchmark strategy for a particular stock, during the trading period. A more detailed explanation of our sector and stock selection process is provided below.

2.1 Random Forest (XOM, CVX & CAT)

Rationale:

Commodity stocks are heavily influenced by the performance of the underlying commodities they are associated with. For example, crude oil-related activities accounted for ~57% of ExxonMobil's net income and ~81.6% of Chevron's net income in 2023. Similarly, commodity-driven operations contributed ~61% of Caterpillar's net revenue in 2023. Given this strong dependence, we believed that incorporating a crude oil index and a broader commodity index would effectively capture the fundamental drivers of these companies. Additionally, because commodities are inherently cyclical, we considered trend-following momentum indicators to be well-suited for modeling their price behavior.

We selected Random Forest for modelling commodity stocks due to its ensemble-based architecture, which combines the predictions of numerous decision trees to enhance accuracy and reduce overfitting. Additionally, Random Forests provide a built-in mechanism for evaluating feature importance, which was particularly useful in our case. Given that commodity stock performance may be influenced by a diverse set of variables — including macroeconomic indicators, commodity indices, and momentum signals — this allows us to identify and prioritise the most informative features for the model, ultimately improving both interpretability and performance.

Feature Engineering Approach:

To capture the underlying trends in commodity prices, we selected the S&P GSCI Crude Oil Index and the Bloomberg Commodity Index as input features, both of which are widely recognised benchmarks in the commodities market. In addition, we incorporated the Moving Average Convergence Divergence (MACD) indicator, a widely used trend-following momentum indicator, to better model the cyclical behavior of commodity stocks. Specifically, we used three components of the MACD:

1. MACD Line (lagged by 2 days):
 - The difference between the 12-day and 26-day exponential moving averages (EMAs), which reflects short- to medium-term momentum shifts.
2. MACD Signal Line (lagged by 2 days):
 - A 9-day EMA of the MACD Line, used to generate buy or sell signals based on crossovers.
3. MACD Histogram (lagged by 2 days):
 - The difference between the MACD Line and the Signal Line

To avoid future bias, we lagged all features (external price index + MACD indicators) by 2 days before incorporating them into our Random Forest model. However, for ExxonMobil, we applied a shorter 1-day lag to the external crude oil index instead of 2, as ExxonMobil tends to respond more quickly to changes in crude oil prices. This is due to ExxonMobil's larger market exposure, higher trading liquidity, and direct operational sensitivity to crude oil fluctuations, which result in faster price adjustments compared to other commodity stocks.

We utilised the parameters n_estimators=100, random_state=42, n_jobs=-1 for our Random Forest model because they strike a good balance between performance, reproducibility, and computational efficiency. Setting n_estimators=100 is the default parameter, and it allows the model to aggregate predictions from a sufficiently large number of trees, improving accuracy and reducing overfitting. The random_state=42 ensures that results are reproducible across runs, which is important for consistent evaluation and debugging. n_jobs=-1 enables the model to use all available CPU cores during training, significantly speeding up computation time, especially when working with large datasets or many trees.

Entry & Exit Rules:

We defined our strategy to enter and exit positions on a daily basis, taking either a long or short position at the T-1 closing price based on the model's predicted stock direction.

1. If the predicted direction is +1, we take a long position
2. If the predicted direction is -1, we take a short position.

In both cases, the position is closed at the T closing price.

Strategy vs. Benchmark

The plots below show the cumulative returns achieved by the Random Forest strategy compared to a simple buy-and-hold benchmark for XOM, CVX, and CAT stocks during the backtesting period (1 March 2024 to 16 January 2025):

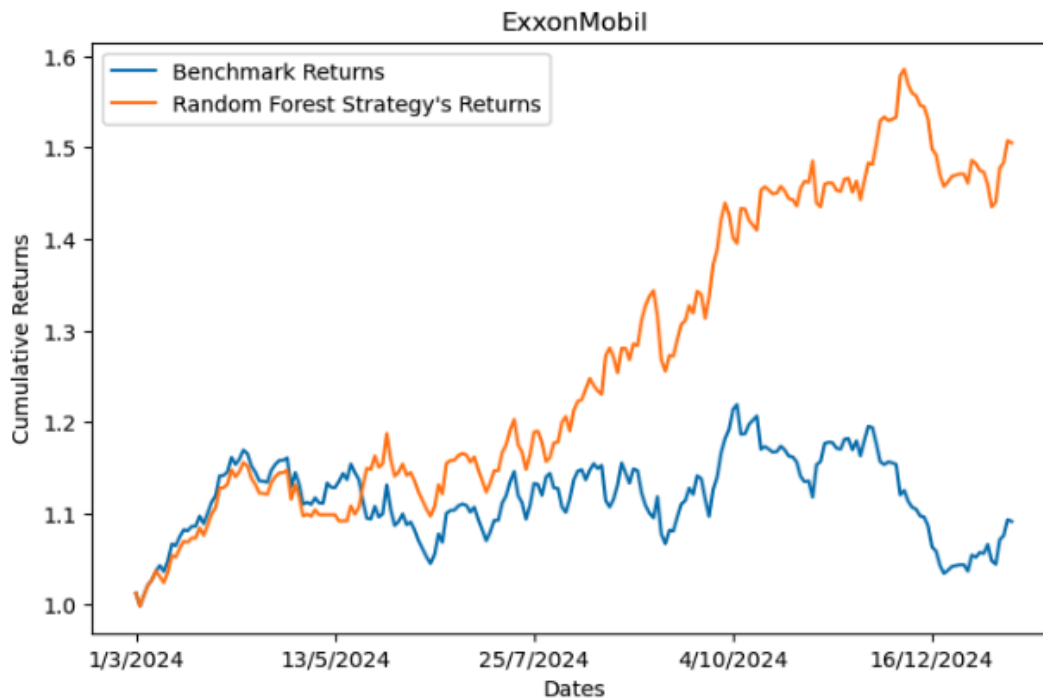


Figure 1: Cumulative Returns of ExxonMobil against Benchmark

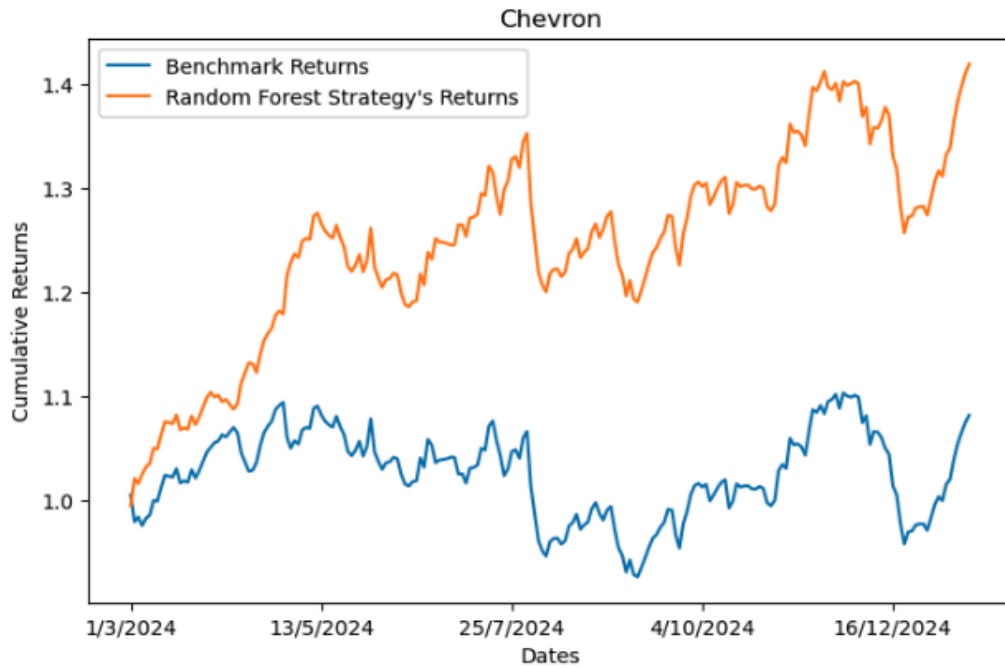


Figure 2: Cumulative Returns of Chevron against Benchmark

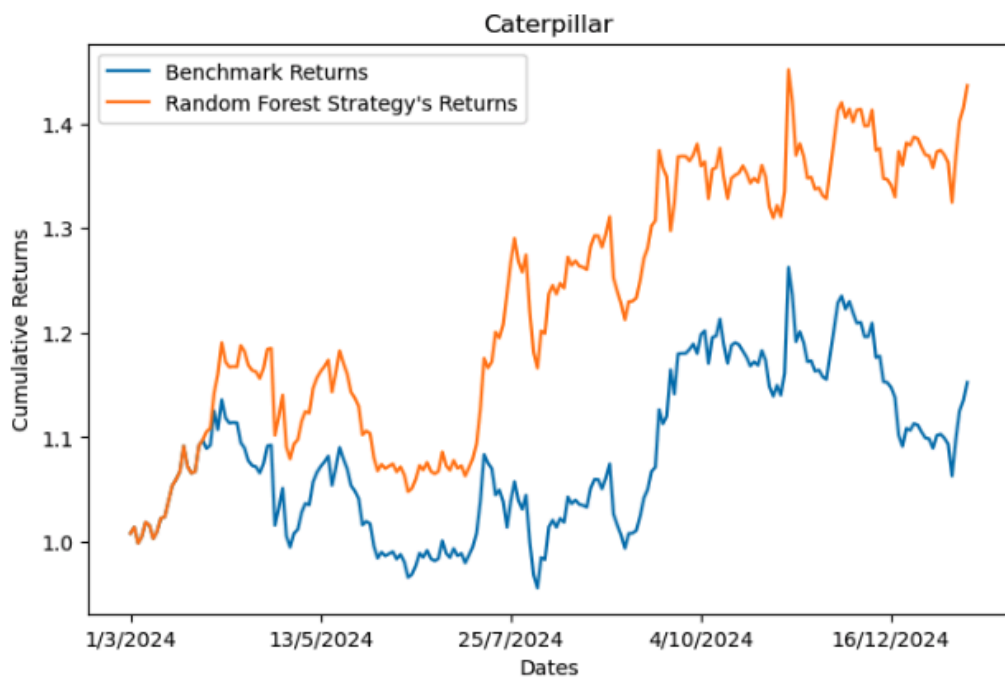


Figure 3: Cumulative Returns of Caterpillar against Benchmark

Stock	Benchmark Returns (2 d.p)	Strategy Returns (2 d.p)	Outperformance (2 d.p)
ExxonMobil	1.09	1.51	+42%
Chevron	1.08	1.42	+34%
Caterpillar	1.15	1.44	+29%

For ExxonMobil, Chevron and Caterpillar, the Random Forest strategy outperformed the benchmark by a total of **~42%**, **~34%** and **~29%**.

For ExxonMobil, while both the strategy and the benchmark showed similar performance in the early months, a clear divergence began to emerge around mid-year. During that time, the benchmark returns began to flatten and decline slightly, whereas the strategy continued to generate upward momentum, significantly widening the return gap. This outperformance can be attributed to the strategy's ability to capture short-term trends and avoid drawdowns that affected the benchmark, particularly in the second half of the year. Even in periods of increased volatility or sideways market movement, the strategy maintained consistent positive returns, highlighting its effectiveness in adapting to varying market conditions.

For Chevron, from the beginning, the strategy consistently outpaced the benchmark, widening the performance gap as the period progressed. While the benchmark returns remained relatively flat or volatile in several stretches, particularly between 03/2024 and 08/2024, the strategy steadily captured upward momentum, demonstrating resilience during periods of uncertainty. This suggests that the model was effective in identifying short-term directional signals and capitalising on trends, even when the overall market was not strongly directional.

For Caterpillar, while both the strategy and the benchmark followed a somewhat similar overall trajectory in the first few months, the strategy began to show clearer separation in performance starting mid-year. During periods where the benchmark exhibited sideways movement or experienced drawdowns, particularly between 05/2024 and 08/2024, the strategy managed to either maintain gains or continue climbing steadily. This indicates that the model was effective in filtering out noise and capturing directional moves, even in choppy market conditions.

In summary, Random Forest consistently outperformed the benchmark for the commodity stocks, ExxonMobil, Chevron, and Caterpillar, by effectively capturing short-term trends and adapting to

various market conditions. Its ability to maintain strong returns even during sideways or volatile periods highlights its robustness and directional accuracy.

2.2 Walk-Forward Decision Tree (JNJ, UNH & NVDA)

Rationale:

A walk-forward decision tree is a decision tree model that's repeatedly retrained on all available historical data up to each point. We used a walk-forward decision tree because, qualitatively, healthcare and semiconductor stocks trade in distinct regimes, such as volatile spikes around earnings or policy news, then quieter drift, making static models stale. For example, on April 17, 2025, UnitedHealth shares plunged over 20% after an earnings-and-guidance shock (Roy & Satija, 2025), triggering a sector-wide sell-off that a daily-retrained model can adapt to in real time.

Decision trees create nonlinear, rule-based splits on engineered features (RSI, MACD, volume surge), capturing conditional “if ... then ...” triggers. For example, “if $RSI > 75$ and volume surge $> 30\%$, then exit long”. Their interpretability lets us inspect branches and confirm that the model is responding to economically sensible signals rather than noise.

Finally, the walk-forward framework replicates live trading: the model only ever sees data up to each “today”. This eliminates look-ahead bias, and retrains daily so parameters never go stale when, for instance, NVIDIA shifts from consolidation into a breakout after an earnings beat.

Feature Engineering Approach:

1. 14-Day RSI
 - A 14-day RSI smooths out two weeks of price action, filtering out noise but remaining sensitive enough for medium-term momentum in relatively stable healthcare names.
2. MACD (12–26 EMA difference)
 - MACD captures crossovers in medium vs. longer-term trends—helpful during volatile weeks (Aryafin, 2025) when a stock shifts from consolidation into a trending move after a clinical trial result or policy announcement.
3. Volume change vs. 14-day average

- Calculating the ratio of volume change against a 14-day average confirms that price moves reflect genuine institutional flows, not just thin-volume noise—surging volume often precedes or accompanies genuine institutional buying/selling in large-cap healthcare names, reducing false breakouts. This is especially important around Nvidia’s peak-volume events (e.g., investor day, product unveilings).

Our decision tree examines possible thresholds on RSI, MACD, and volume change relative to a 14-day average, choosing the split at each node that is best at separating days with positive next-day returns from those with non-positive returns. It repeats this recursively, growing a small tree of maximum depth = 3, whose leaves contain mostly ‘up’ days or ‘down’ days.

For each candidate threshold, it partitions the training examples into those with feature $\leq$ threshold and feature $\geq$ threshold. Then, the Gini impurity is computed in each child node. These impurities are weighed by the number of samples on each side, and then subtracted from the parent’s impurity to calculate the impurity drop.

The algorithm compares all candidate splits across the features RSI, MACD, and Volume Change, and chooses the one that yields the largest impurity reduction.

Entry & Exit Rules:

We defined our strategy to enter and exit positions on a daily basis, taking either a long or short position at the T-1 closing price based on the model’s predicted stock direction.

3. If the predicted direction is +1, we take a long position
4. If the predicted direction is -1, we take a short position.

In both cases, the position is closed at the T closing price.

Strategy vs. Benchmark

The plots below show the cumulative returns achieved by the Walk-Forward Decision Tree strategy compared to a simple buy-and-hold benchmark for JNJ, UNH, and NVDA stocks during the backtesting period (1 March 2024 to 16 January 2025):

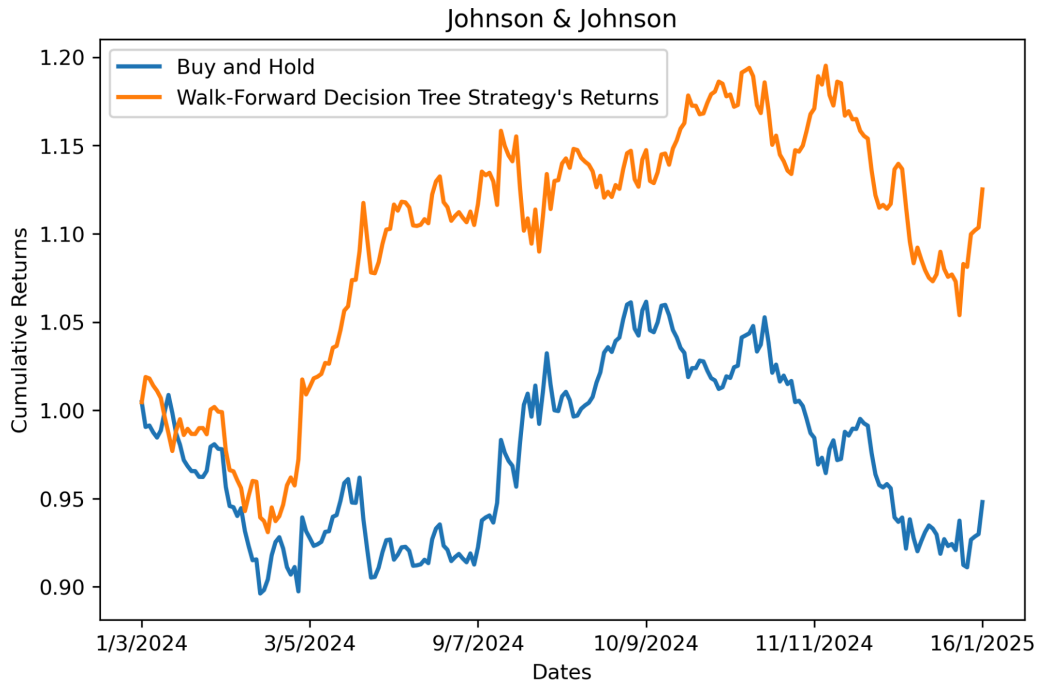


Figure 4: Cumulative Returns of Johnson & Johnson against Benchmark

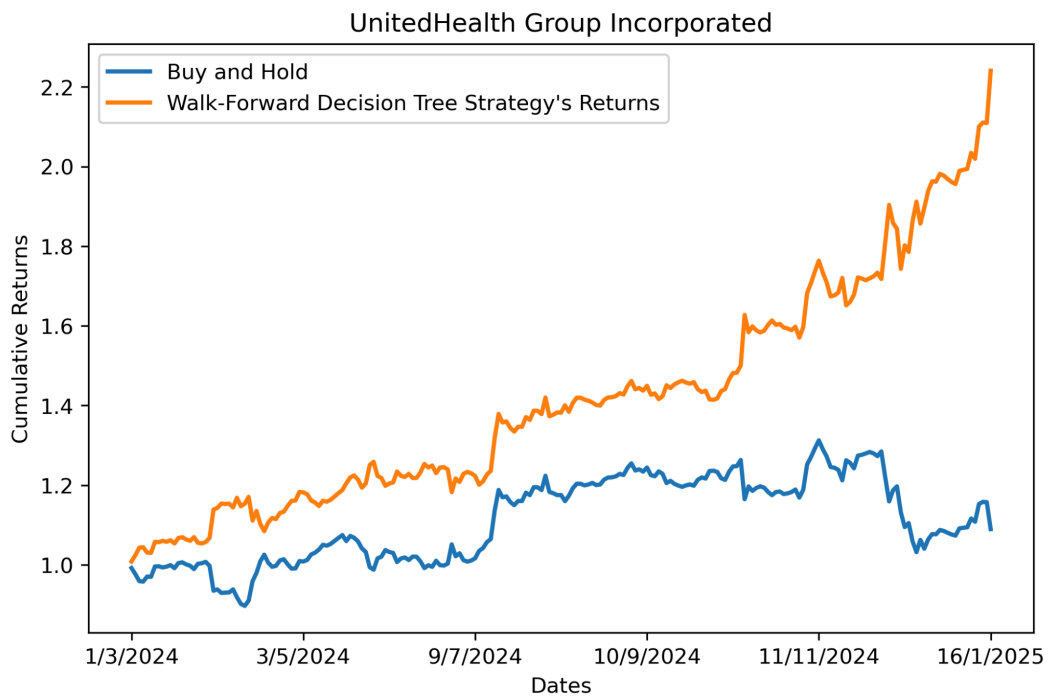


Figure 5: Cumulative Returns of UnitedHealth Group Incorporated against Benchmark

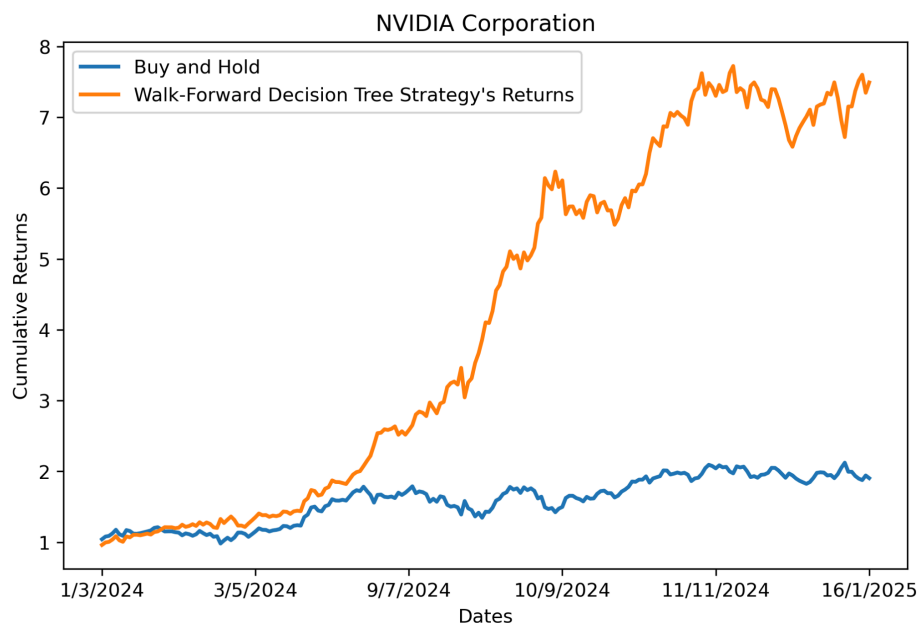


Figure 6: Cumulative Returns of NVIDIA Corporation against Benchmark

Stock	Benchmark Returns (2 d.p)	Strategy Returns (2 d.p)	Outperformance (2 d.p)
Johnson & Johnson	0.95	1.13	+18%
UnitedHealth Group Incorporated	1.09	2.24	+115%
NVIDIA Corporation	1.90	7.50	+559%

For Johnson & Johnson, UnitedHealth Group Incorporated and NVIDIA Corporation, our strategy outperformed the benchmark (buy and hold) by **~18%**, **~115%** and **~559%** respectively.

For Johnson & Johnson, the strategy never fell as far as the benchmark in early March, then quickly recovered on oversold RSI and volume-surge signals. It increased by late summer, sheltered roughly half of the November–December drawdown, and finished the period at **~+12.5%** while buy-and-hold languished at **~ -5.2%**

For UnitedHealth Group, the strategy held above 1.00 through the March dip while buy-and-hold dipped, then captured every major upswing—peaking before November and surging to **+124%** by mid-January. By contrast, buy-and-hold only climbed modestly pre-November before ending at **~ +9%**.

For NVIDIA Corporation, both curves started at parity, but after September’s MACD-driven breakout the strategy vaulted above and into the 6–7 range, peaking near **~ +650%** by January

16, 2025. Buy-and-hold, in comparison, rose steadily to only +90 %. The decision-tree's ability to enter on strong momentum signals and exit on volume surges transformed a ten-month period of ~ 90% benchmark gains into over ~ 650% strategy returns.

2.3 Linear Regression (AAPL & MSFT)

Rationale:

Tech stocks are known for their high volatility, driven by factors such as rapid innovation cycles, shifting consumer trends and market speculations. This volatility adds considerable noise to their price movements, resulting in challenging predictions. Linear regression offers a way to simplify this complex behavior by modeling the relationship between a stock's future price and relevant factors like past prices and their moving averages, using a simple mathematical function that is linear in terms of the feature's coefficients. By assuming a linear relationship, it helps filter out some randomness and identifies underlying trends or patterns hidden in the data. Moreover, the features can be modified, in an attempt to capture further intricacies of market behavior, for instance by pre-composing the features with nonlinear mathematical functions. Lastly, linear regression provides a useful starting point for understanding and forecasting tech stock behaviours in an interpretable and constructive way. As an example, by reflecting on the coefficients of the features obtained from training, it allows us to analyse the significance of the respective features in predicting future prices.

Feature Engineering Approach:

The two main features we chose to work with are the Closing Price of previous days and their exponential moving average (EMA). In particular,

1. Lagged Closing Prices (1-day and 2-day)
 - The main purpose of the Lagged Closing Prices was to serve as a firm base in predicting future Closing Prices
2. Lagged Exponential Moving Average (Window Size = 20, Alpha = 0.3) (1-day & 2-day)
 - A low alpha, such as 0.3, places less weight on recent observations, smoothing out short-term fluctuations while still adapting to recent market behavior—this is especially important given the high volatility of tech stocks.
 - A window size around 20 strikes a balance between being short enough to remain responsive to recent price changes and long enough to filter out noise, making it well-suited for capturing meaningful trends in volatile tech stocks.

We worked with both of these features to balance responsiveness and stability in capturing price trends. We then further made use of squared and square-rooted EMA values to introduce nonlinearity into the model without increasing its complexity too much, allowing it to potentially capture subtle shifts in momentum that linear combinations might miss.

While we initially posited that stocks within the same sector would exhibit similar price patterns, after extensive testing, we found that no matter how we tuned the parameters, Apple and Microsoft consistently outperformed their benchmark by a significantly larger margin compared to Google. This divergence suggested that a single model might not generalize well across all three. To avoid time inefficiencies, we dropped Google and worked with Apple and Microsoft only.

Moreover, we also observed that certain features which enhanced prediction accuracy and return for Apple did not translate as well to Microsoft, and vice versa. For example, Apple responded better with Closing Price as features, while Microsoft's performance improved without them. As a result, we decided to develop two tailored strategies, one for Apple and one for Microsoft, each personalized with distinct feature sets to maximize predictive power and trading performance.

The final decision to use specific values, such as an alpha of 0.3 and window size of 20 were informed by empirical testing and cross-validation. We compared different alpha values and window sizes, assessing model performance based on metrics like mean squared error and overall return by the strategy. Ultimately, the chosen parameters provided a good trade-off between capturing meaningful trends and avoiding overfitting on noisy data.

With the relevant final features in place, we trained our model dynamically, on each day prior to every trading day. This allowed us to make use of the most updated changes to our features before entering on a trade.

The trained model was then used to predict the Closing Price of the next trading day. With this prediction, we were able to make use of today's Closing Price to measure our predicted return for tomorrow. Based on the signs of the predicted returns, we converted it to either +1, indicating a positive predicted direction of return, or -1, indicating a negative predicted direction of return.

Entry & Exit Rules:

We defined our strategy to enter and exit positions on a daily basis, taking either a long or short position at the T-1 closing price based on the model's predicted stock direction.

5. If the predicted direction is +1, we take a long position
6. If the predicted direction is -1, we take a short position.

In both cases, the position is closed at the T closing price.

Strategy vs. Benchmark:

The plots below show the cumulative returns achieved by the two different Linear Regression Strategies (for each of the stocks respectively) compared to a simple buy-and-hold benchmark for Apple (Figure 7) and Microsoft (Figure 8) during the trading period.

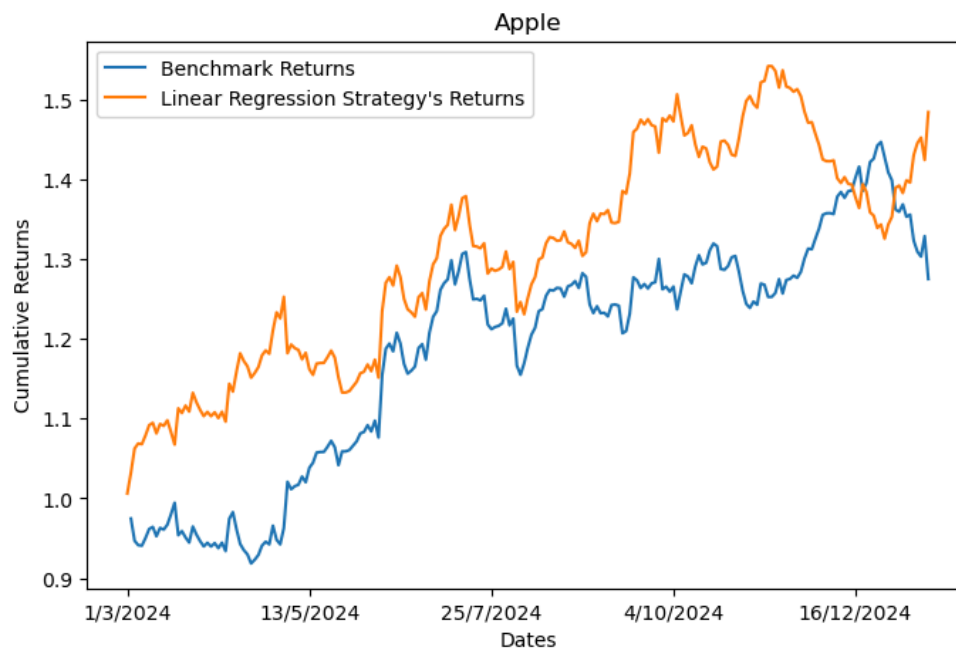


Figure 7: Cumulative Returns of Apple against Benchmark

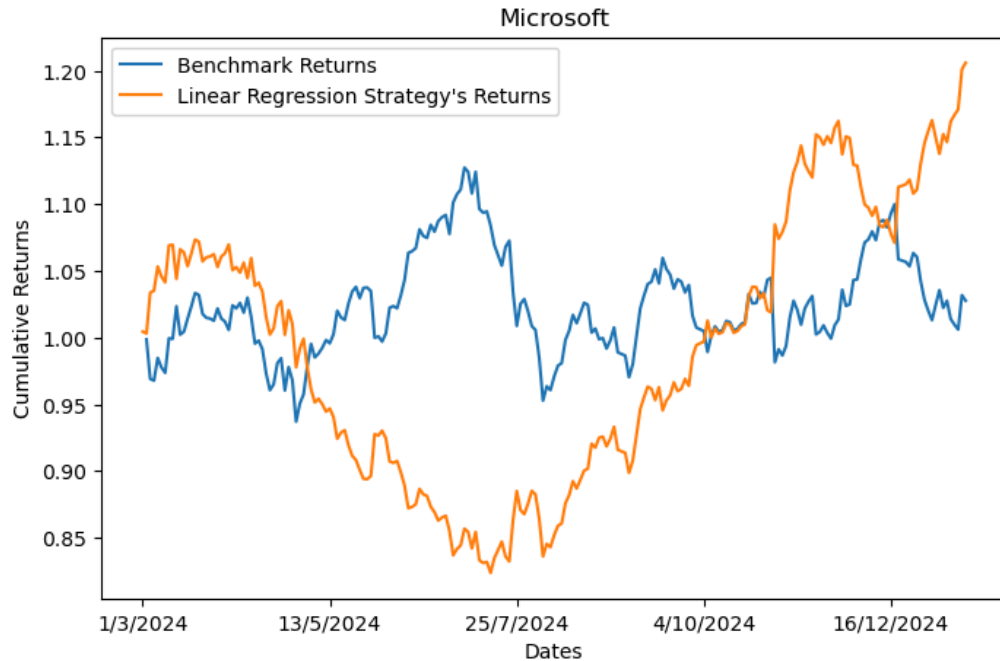


Figure 8: Cumulative Returns of Microsoft against Benchmark

Stock	Benchmark Returns (2 d.p)	Strategy Returns (2 d.p)	Outperformance (2 d.p)
Apple	1.28	1.48	+20%
Microsoft	1.03	1.21	+18%

For Apple and Microsoft, our strategy outperformed the benchmark (buy and hold) by **20% and 18%** respectively.

For Apple, the cumulative returns generated by the linear regression strategy were consistently higher than the benchmark throughout most of the trading period. On a general level, it can be observed that Apple's closing prices tend to exhibit a positive upward drift over time. This persistent directional trend provides an ideal environment for linear regression, where the model is then able to capture and extrapolate the upward momentum effectively. The consistency in Apple's price behavior reduces the noise and volatility that often disrupt model performance, enabling the strategy to outperform the benchmark more reliably.

Although there was a slight dip toward the end, the strategy rebounded sharply and ultimately surpassed the benchmark.

In contrast, the cumulative returns of the Microsoft strategy lagged behind the benchmark for a significant portion of the trading period, particularly around the midpoint. This may be attributed to the oscillating nature of Microsoft's Closing Price in the first half, where the closing prices generally fluctuated around a stable average and moved without any statistical drift. Without a directional trend to anchor onto, linear regression tends to struggle, resulting in the model's prediction to be misaligned with actual movements, leading to underperformance during periods of price consolidation.

However, in the latter half, the strategy experienced a strong upward trend, eventually overtaking the benchmark by the end.

Upon further analysis, we also uncovered an aspect worth mentioning.

Particularly, there are frequent episodes of exaggerations in prediction. For example, consider Figure 9

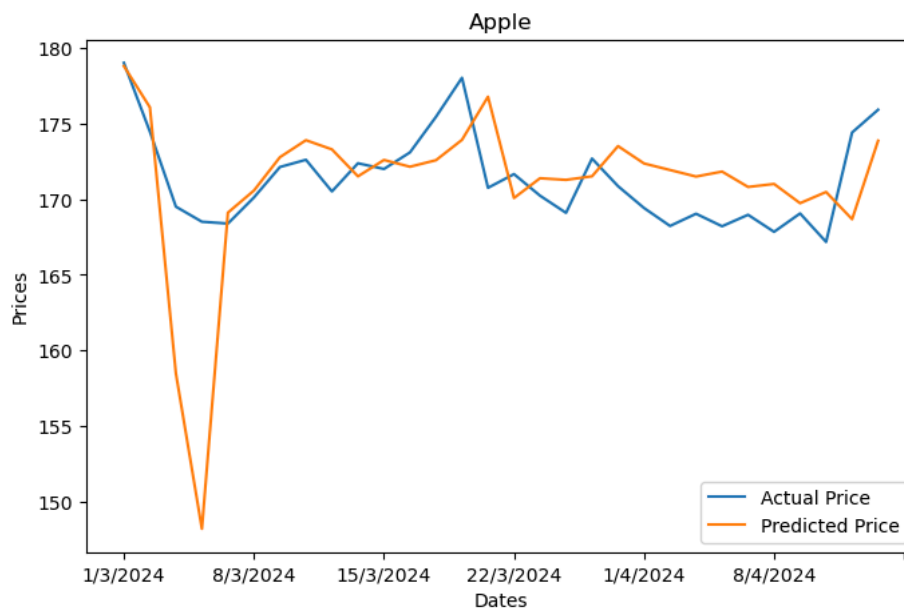


Figure 9: Actual and Predicted Closing Prices of Apple

The figure illustrates how the model's predicted closing prices align with the actual closing prices. Notably, between 1st March 2024 and 8th March 2024, the predicted prices fall significantly below the actual prices. Such deviations, where the model underestimates or overestimates market movements, can have a mixed impact on performance. While they may sometimes capture turning points early and enhance returns, they also risk generating inaccurate signals that lead to suboptimal trades, thereby diminishing overall strategy effectiveness.

2.4 MLP Classifier (NKE & BA)

Rationale:

Nike and Boeing's stock prices are driven by a combination of macroeconomic shifts and sector-specific risks, resulting in complex and nonlinear return behaviors. Quantitatively, both stocks exhibit volatility clustering, and frequent trend reversals, particularly during earnings announcements or geopolitical events. Qualitatively, Nike's exposure to consumer sentiment and Boeing's sensitivity to regulatory and geopolitical factors suggest regime-switching dynamics that simple models struggle to capture. Capturing these dynamics required a flexible modeling framework, for which a Multi-Layer Perceptron (MLPClassifier) was selected to model hidden nonlinear relationships between lagged returns, momentum indicators, and trend signals, allowing for dynamic adaptation to evolving market conditions.

The MLPClassifier model was configured with two hidden layers (64 and 32 neurons) and was trained on the engineered features. To prevent overfitting, regularization was applied (`alpha = 0.01`) along with early stopping, which halts training once performance stops improving. Additionally, a fixed random state (`random_state=1`) was set to ensure that results are predictable and reproducible across multiple runs. This setup allowed the model to effectively learn and capture the nonlinear momentum and trend behaviors characteristic of Nike and Boeing stock returns.

Feature Engineering Approach:

To effectively model short-term momentum and broader trend shifts for Nike and Boeing, the following features were constructed:

1. Lagged Returns (1-day and 2-day)
 - To capture immediate price momentum changes
2. Exponential Weighted Moving Averages (EWMAs) of Delta Returns
 - Calculated using spans of 3 and 5, to model different speeds of trend formation while emphasizing recent price changes
3. Lagged EWMA Features
 - Where all EWMA signals were shifted by one day, ensuring that only past information is used during prediction and fully avoiding lookahead bias
4. Interaction Terms
 - Lagged Returns multiplied by Lagged EWMA were created to allow the model to capture nonlinear effects, such as when recent momentum interacts with recent trend strength
5. Delta Returns (lagged)

- Included to provide additional momentum-based insights

Before model training, all features were standardized using *StandardScaler* to ensure faster convergence and more stable optimization.

Additionally, during training, early stopping was enabled to prevent overfitting, allowing the model to halt training automatically if performance no longer improved.

The model is retrained dynamically every 10 trading days based on all available data up to that point, ensuring that the MLPClassifier continually adapts to new market information as conditions evolve.

Entry & Exit Rules:

We defined our strategy to enter and exit positions on a daily basis, taking either a long or short position at the T-1 closing price based on the model's predicted stock direction.

7. If the predicted direction is +1, we take a long position
8. If the predicted direction is -1, we take a short position.

In both cases, the position is closed at the T closing price.

Strategy vs. Benchmark

The plots below show the cumulative returns achieved by the MLP Classifier Strategy compared to a simple buy-and-hold benchmark for Nike (Figure 9) and Boeing (Figure 10) during the trading period in the next page.

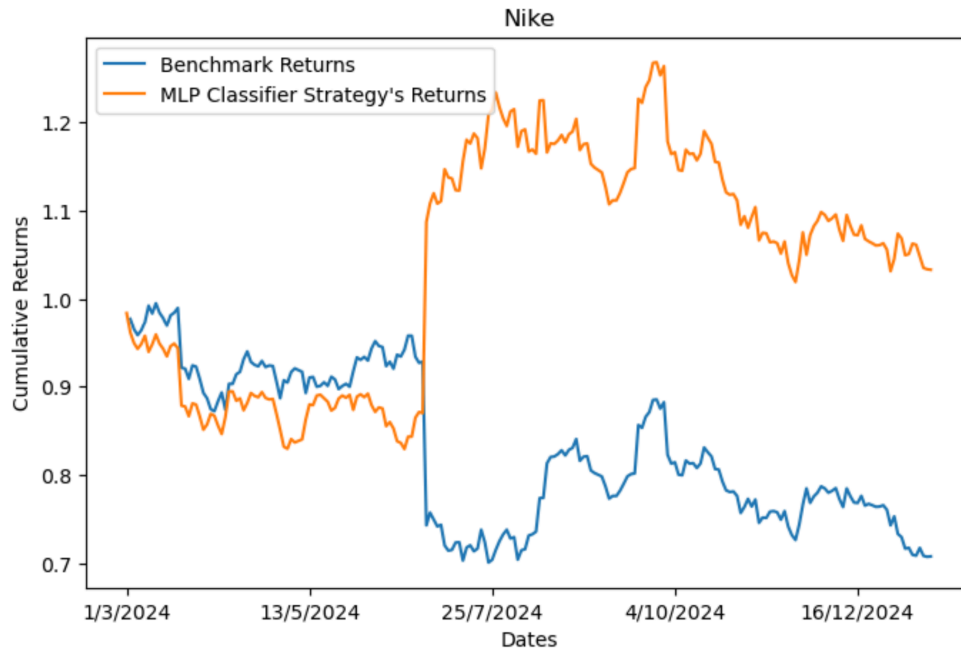


Figure 10: Cumulative Returns of Nike against Benchmark

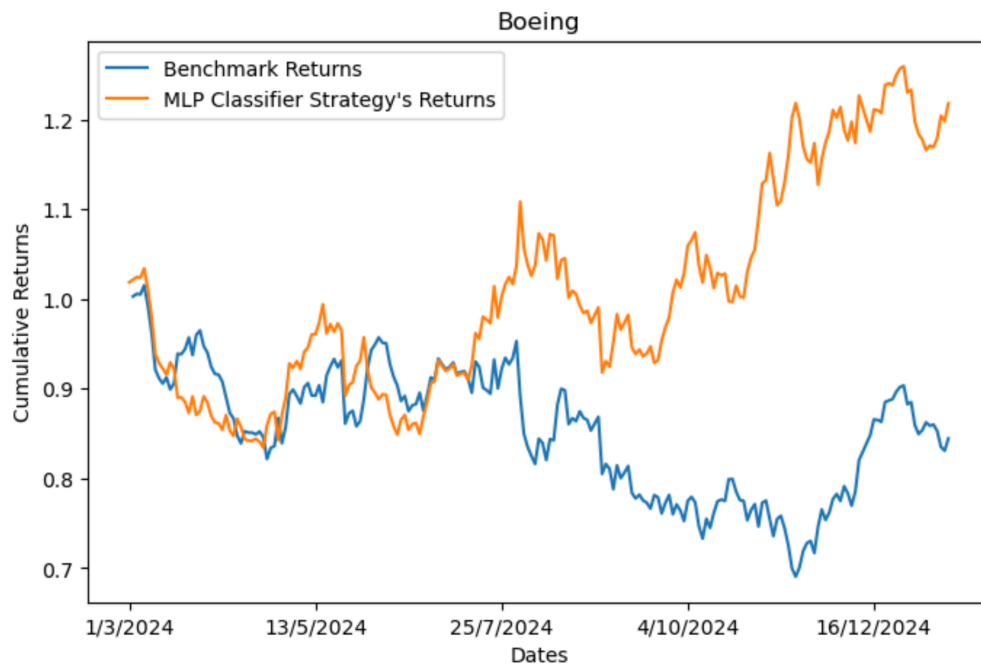


Figure 11: Cumulative Returns of Boeing against Benchmark

Stock	Benchmark Returns (2 d.p)	Strategy Returns (2 d.p)	Outperformance (2 d.p)
Nike	0.71	1.03	+32%
Boeing	0.84	1.22	+38%

For Nike and Boeing, our strategy outperformed the benchmark (buy and hold) by **32% and 38%** respectively.

For Nike, the benchmark returns for Nike declined steadily over the year, ending significantly below the starting point. In contrast, the MLP Classifier strategy achieved strong cumulative growth, clearly outperforming the benchmark. Around mid-2024, a sharp breakout in cumulative returns was observed, indicating that the model effectively captured major price swings and reversed downtrends that the benchmark failed to avoid.

Similarly, Boeing's benchmark returns faced a prolonged decline, reflecting a tough and volatile market environment. The MLP Classifier strategy not only navigated this volatility but also consistently outperformed the benchmark, achieving substantial cumulative gains. The strategy demonstrated notable resilience during periods of heightened market turbulence, maintaining higher cumulative returns with smaller drawdowns relative to the benchmark.

Overall, the MLP Classifier worked well for volatile, mean-reverting stocks where the benchmark strategy underperformed. However, it underperformed on high-momentum stocks like Tesla and Nvidia, where strong persistent trends were often misread as short-term noise.

While the overall strategy remained profitable, the prediction accuracy for both Nike and Boeing hovered around 50%, indicating that a handful of well-timed, high-impact trades drove the majority of cumulative returns. This highlights that although the model captured major reversals effectively, the underlying signal quality was inconsistent.

Despite this, the MLPClassifier demonstrated strong trend-capture abilities, avoiding major drawdowns and steadily compounding returns through volatile market phases. That said, during periods of low volatility or sideways price action, the model's performance became more erratic, with false signals occasionally leading to short-term losses.

Section 3: Strategy Performance

Entry & Exit Rules:

We defined our portfolio strategy to enter and exit positions on a daily basis, taking either a long or short position at the T-1 closing price based on our model's predicted stock direction.

1. If the predicted direction is +1, we take a long position
2. If the predicted direction is -1, we take a short position.

In both cases, the position is closed at the T closing price.

Capital Allocation:

For our capital allocation strategy, we employed the Kelly fraction, $\frac{\mu - r}{\sigma^2}$,

where

μ is the annualised mean log return of a stock,

σ is the annualised standard deviation of log returns of a stock &

r is the risk-free rate

For each of the 10 selected stocks, we computed the annualised mean and standard deviations of log returns on a daily basis (for each trading day), by using a rolling object of size 21 on the historical log return values. We then made use of the above mentioned formula to compute the optimal fraction.

We aim to use the obtained fraction computed for each stock (for each trading day) to represent the proportion of our **entire current capital** that is to be invested in that particular stock on that particular day.

However, a few adjustments are needed. Firstly, the above formula does not ensure non-negativity. Next, it also does not limit the proportion of our entire capital to be allocated in any stock. For example, it might even be the case where we obtain a proportion of 1 for every stock, resulting us to incur a 10x leverage on our entire portfolio. Lastly, the formula also leaves the possibility of incurring a huge proportion of uninvested capital, which can significantly hinder our gains. To circumvent this, we embark on a two-step scaling.

First, we used a *MinMaxScaler* with a feature range of $(0, 0.1)$, ensuring that each Kelly fraction is non-negative and the total Kelly fraction summed across all 10 stocks would never exceed 1,

preventing leverage and overall allocation beyond the available capital. This also guaranteed no single stock would receive a full allocation (i.e., a weight of 1).

Second, for each trading day, we normalised each stocks's scaled Kelly fraction using the formula:

$$\frac{\text{Daily Stock Kelly}}{\text{Total Daily Kelly}}$$

where

Total Daily Kelly is the sum of the scaled Kelly fractions for that day across the 10 stocks

This scaling ensures that the sum of *Daily Stock Kelly* across 10 stocks will always sum up to 1, ensuring full capital allocation, effectively redistributing any unallocated capital proportionally among the selected stocks. With this, we obtain the final scaled Kelly fraction for each stock on a particular day.

We then determine each stock's daily capital allocation for trading by multiplying each stock's final scaled Kelly fraction by the total equity available on that day. On the first day, the total equity is \$50,000, which represents our initial capital. However, this value is updated daily as we realise profits and losses from our trades, meaning our equity and thus our allocations evolve over time.

However, in accordance with the assignment guidelines, we ensured that on any given day, if a stock is shorted, only half of the allocated capital is used to execute the short position when calculating profit and loss.

3.1 Portfolio Equity vs. Benchmark Equity

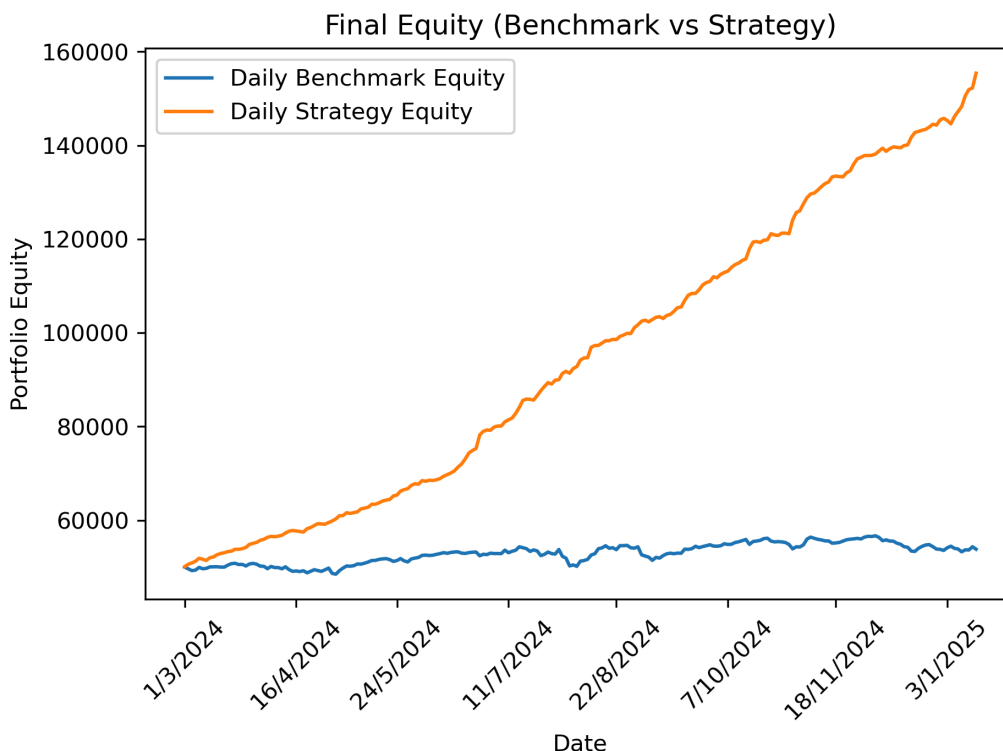


Figure 12: Cumulative Returns for Final Equity

Portfolio Performance vs. Buy-and-Hold (Mar 1, 2024–Jan 16, 2025)

In Figure 10, we compare two \$50,000 portfolios over the ten-month period from March 1, 2024 to January 16, 2025. The blue legend is a benchmark buy-and-hold strategy, equally allocating up to \$5,000 per stock across ten tickers. The orange legend is our “hybrid machine-learning” strategy, which employs Linear Regression (AAPL, MSFT), Random Forests (CVX, XOM, CAT), MLP classification (NKE, BA), and Decision Trees (JNJ, UNH, NVDA). The benchmark returned a final portfolio equity of ~\$53,841, whereas our strategy returned a final portfolio equity of ~\$155,419, outperforming the benchmark by \$101,578, or 203%

Overall, the outperformance of our hybrid machine-learning portfolio is clear. Its daily, signal-driven reallocation, tempered by Kelly fraction sizing, allowed it to embrace winners aggressively and exit losers swiftly. The result is a smoother and higher equity curve compared to the benchmark static buy-and-hold, demonstrating better performance across ten stocks.

Section 4: Analysis & Future Improvements

4.1 Analysis of Returns

Across all sectors, our strategies consistently outperformed the buy-and-hold benchmark, achieving almost 3x of the final portfolio value compared to the benchmark strategy.

Specifically:

1. Random Forest (XOM, CVX, CAT)
 - Captured commodity-driven trends very effectively, with outperformance margins of 29% to 42%. The strategy was resilient even during volatile sideways movements, leveraging cyclical momentum well.
2. Walk-Forward Decision Tree (JNJ, UNH, NVDA)
 - Excelled particularly with NVIDIA (559% outperformance). The daily retraining allowed the model to respond to real-time regime shifts, like healthcare earnings shocks and semiconductor breakouts, minimizing drawdowns and maximizing upside capture.
3. Linear Regression (AAPL, MSFT)
 - Simpler models still generated 18%-20% outperformance over benchmarks, especially benefiting from Apple's steady upward drift. Some instability was observed in Microsoft's mean fluctuating price behavior, but later recovery showed adaptability.
4. MLPClassifier (NKE, BA)
 - Designed for volatile and nonlinear stocks, the MLPClassifier generated solid returns (32% outperformance for Nike, 38% outperformance for Boeing) by effectively catching mean-reverting reversals. However, prediction accuracy hovered around 50%, implying that the gains were driven by a handful of critical trades rather than consistently accurate predictions.

Overall, our success hinged on:

1. Adaptive retraining (dynamic models updated every few days)
2. Sector-specific feature engineering (not one-size-fits-all)
3. Dynamic capital allocation (Kelly Criterion adjustment)

4.2 Future Improvements for the Overall Trading Strategy

Moving forward, here are some future improvements we can consider:

4.2.1 Adjusted Price Data for Corporate Actions

To ensure our backtests and live returns are accurate and robust, we can adjust all price series for corporate actions, such as dividends, stock splits, spin-offs. This makes sure that every return calculation reflects economic value rather than solely price changes. At the same time, enriching our dataset with intraday or high-frequency observations such as 5-minute bars would allow us to capture more nuanced patterns (Chakrabarty, 2022) and potentially improve entry and exit timing.

4.2.2 Alternative or unstructured data sources

Other than price and volume indicators, we can enrich our signal set by using additional alternative data sources. For example, monitoring insider-trading filings and unusual spikes in equity or options volume can reveal early signs of institutional positioning (Bookmap, 2024), while broker sentiment surveys and social-media feeds such as Reuters and Twitter offer real-time clues to market psychology (Warkulat et al., 2024). Finally, options-market metrics like implied volatility capture forward-looking uncertainty and tail-risk expectations (Dierkes et al., 2024), rounding out a more comprehensive and anticipatory trading framework.

4.2.3 Portfolio construction and money management

We can also improve risk control by moving beyond the traditional full-Kelly approach. For example, implementing fractional Kelly strategies (such as half-Kelly), capped Kelly methods that impose a maximum position size, or asymmetric Kelly formulations can all help to limit drawdowns and better manage tail risk, striking a better balance between growth and preservation.

4.2.1.4 Strategy Implementation

Going forward, we can improve our implementation by using a systematic approach for allocating any leftover cash after applying the Kelly criterion—ensuring we’re fully invested without leaving idle capital—and by refactoring the data-preparation pipeline so that every stock’s DataFrame is created and standardized from the outset, reducing repetitive code and minimizing the chance for indexing or formatting errors. These changes will not only make our backtests more accurate and efficient but also dramatically simplify future feature additions and strategy experiments.

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